VDK/TLM Creation

Rapid Training

X-2025.06

Module 1: Lab G

Model Usage in a VDK

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Lab Overview

You can integrate models developed or imported in TLM Creator easily into your virtual prototype design using the VDK Creation perspective of Virtualizer Studio (VDK Creator).

Your task for the current exercise to go through the steps to add the timer model (created in the earlier labs) in a VDK project in Virtualizer Studio.

Open the timer model library in the VDK Creator and instantiate it into your virtual prototype design.

Task-1: Open the Timer Project

***Motivation**: We have already created basic version of a programmable interval timer model in the previous labs. We will use this for the current lab -alternatively you can use a provided starting point project that contains a version of the timer model already.*

If you have completed some of the previous lab (Module 1 Lab D or any subsequent lab), you can just continue with your existing TLM Creator project and skip this task and continue with Task-2.

1. If have not completed the previous lab or want to start from a prepared starting point for this lab, use the following steps to create the starting point.

   i. Open a console and invoke Virtualizer studio from the Synopsys Virtual Prototyping installation.

   ```
   % source <installDir>/SLS/windows/setup.bat vi[rtualizer] (Windows)

   % source <installDir>/SLS/linux/setup.csh|sh vi[rtualizer] (Linux)
   ```

   This sourcing option picks up the Virtualizer *Elite* or *Base* license in that order, whichever is available. For more details on other options, refer to the VP documentation.

   i. Go to the `<Lab-data>/data/module_1` directory and start Virtualizer Studio.

   ```
   vs -d start_Module1LabG &
   ```

2. Close the welcome screen and import the starting point TLM Creator Project by clicking ***File*** → ***Import…*** menu option.

   i. In the ***Import*** dialog, select ***General*** → ***Existing Projects into Workspace*** to select what you want to import.

   ii. Click ***Next*** to get to the ***Import Projects*** dialog. Browse to the `timer` directory in the `start_Module1LabG` directory and click ***Finish*** (Make sure you are in the right directory – it's easy to end up in the wrong directory!).

   iii. Select the `timer.tlmc` file in the Project Explorer and click ***Generate Sources Code*** from the ***Build*** toolbar button options. Also, generate the documentation using the ***Generate Documentation*** option.

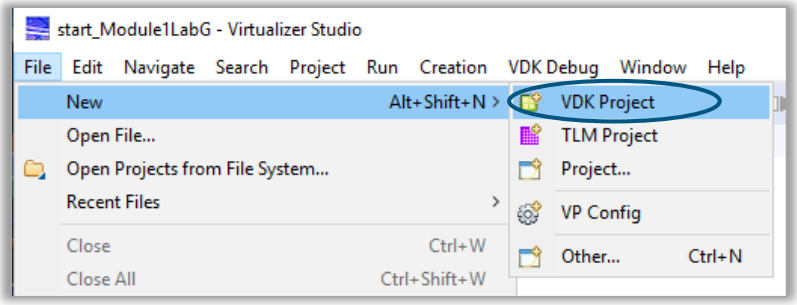
This will provide you with the starting point `timer` model for this lab.

Task-2: Open the *timer* Model Library in VDK Creation Perspective

***Motivation**: We will now use the timer model library in the VDK Creation perspective of Virtualizer Studio and add it to our VDK project.*

3. Click ***File -> New*** menu option and select ***VDK Project*** from the list.

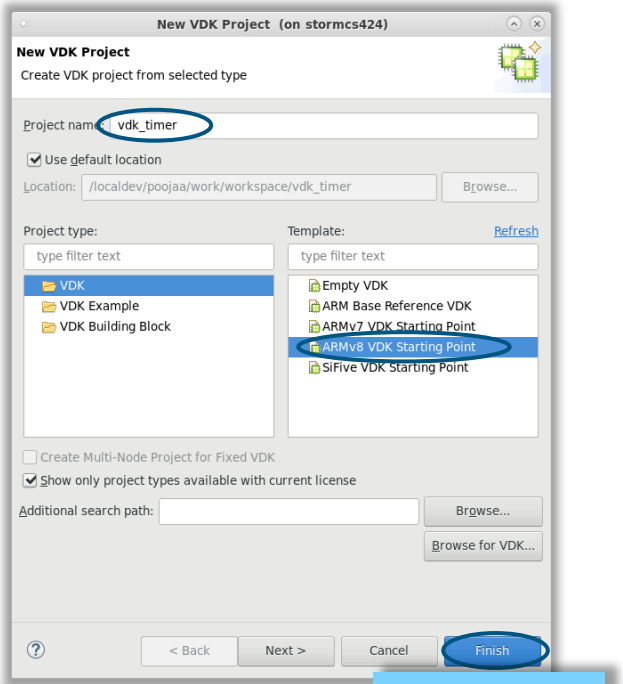
   If you don't see this option, use ***Other…*** and select the option from under the *Virtual Prototyping* group.



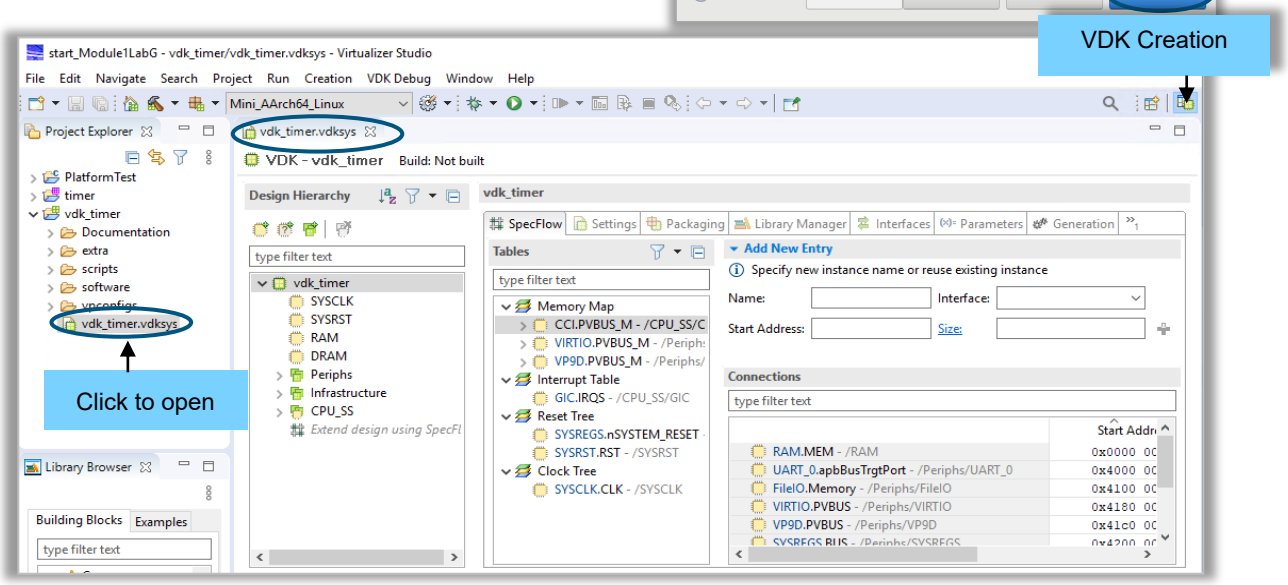
4. The ***New VDK Project*** dialog appears. Specify the following details:

   - Type `vdk_timer` in the ***Name*** field,
   - Select the `ARMv8 VDK Starting Point` as the ***Project type*** to start with,
   - Click ***Finish***.

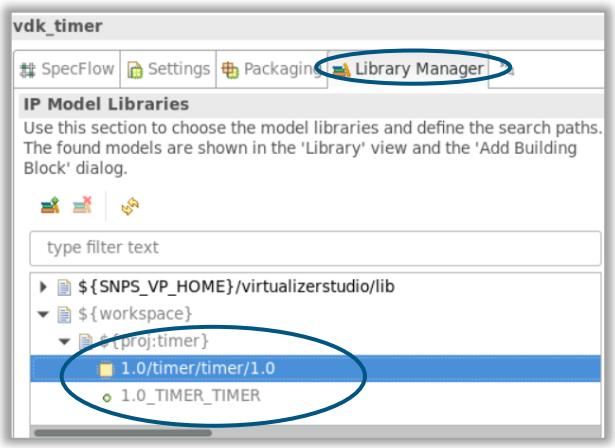
   When prompted, click ***Open Perspective*** to open the ***VDK Creation*** perspective. Your VDK project gets created and opens in the ***VDK Creation*** perspective.



5. Review the components of your project in the ***Project Explorer*** and if not already opened, click to open `vdk_timer.vdksys` in the VDK Creation GUI.



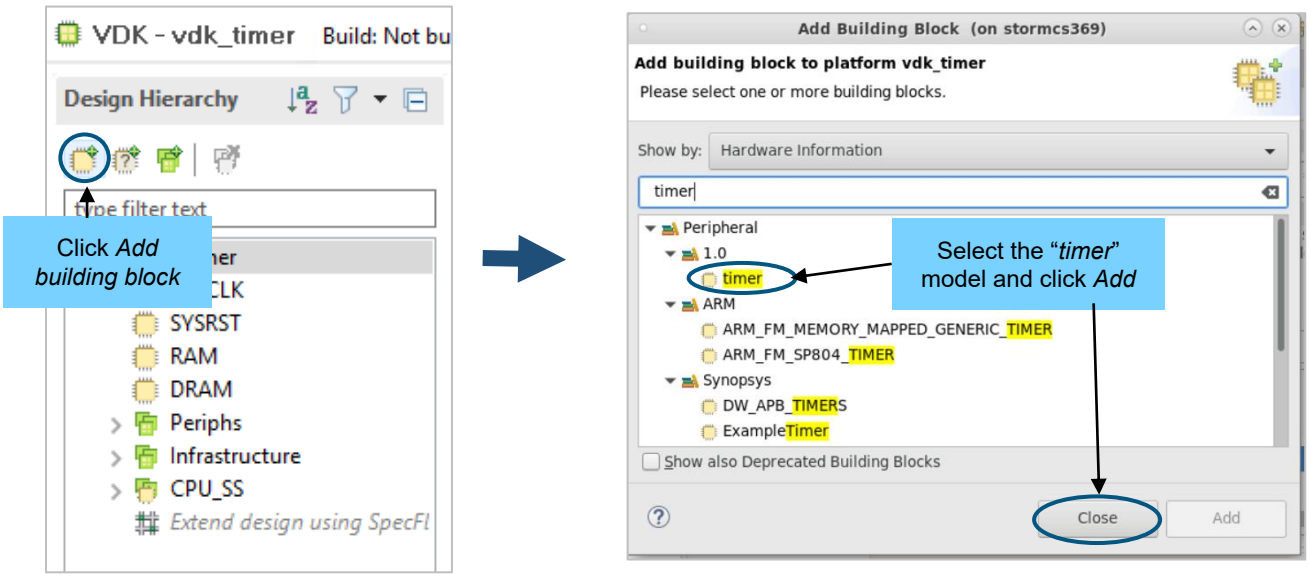
6. Open the ***Library Manager*** tab and search for the `timer` model library reference in your workspace created in the previous labs. Expand to view the details.



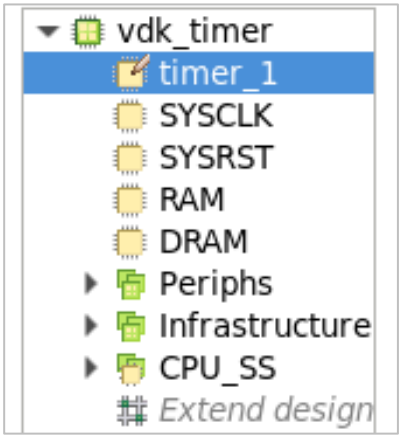
Make sure the timer library is automatically added.

Note: In case, you do not find a reference of the timer model in this tab, you need to manually add the library path so that you can add the component to the VDK using the following steps.

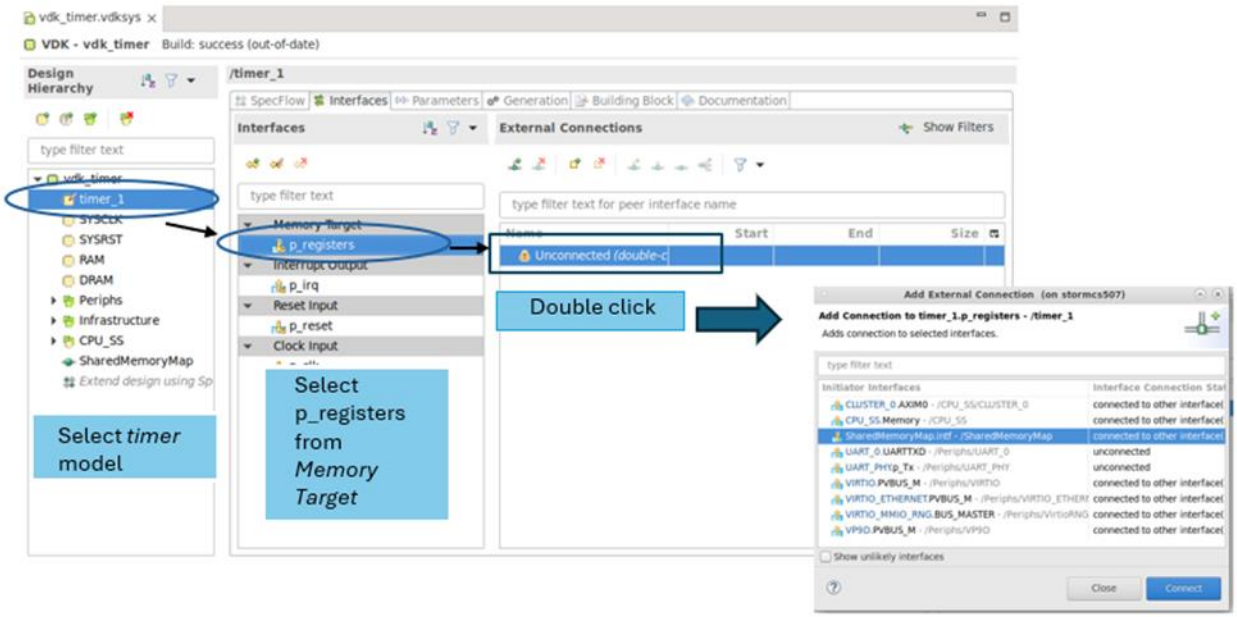
- Right-click in the ***Library Manager*** tab, select the ***Add Model Library*** option
- Browse to the `<yr-wrkspc>/timer/Tool/PCT` path, and select the `timer.xml` library file.
- Focus on the Design Hierarchy view and click the ***Add building block*** button. In the Add dialog, search for "`timer`" model, select the model from your workspace and click ***Add***.



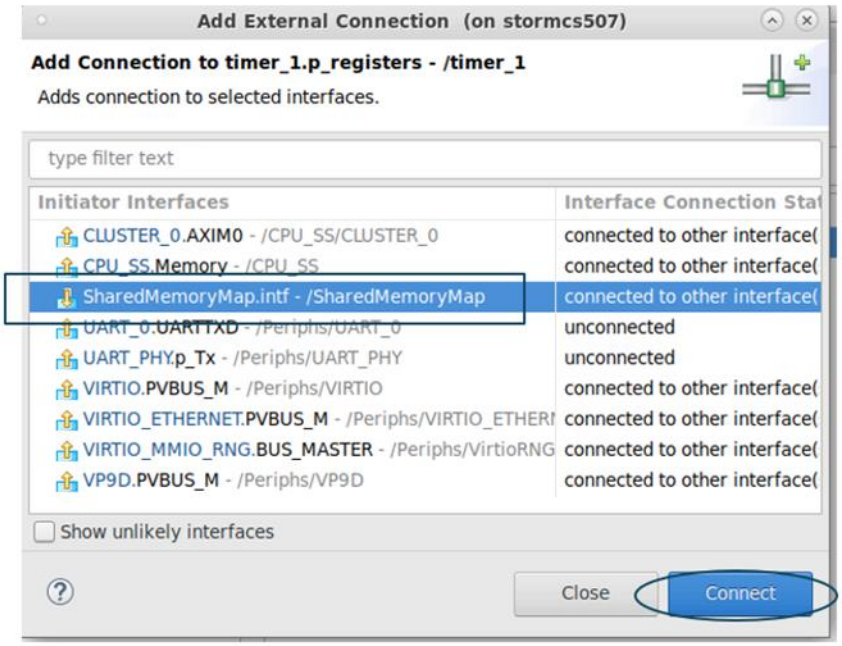
The timer model gets added to the vdk_timer project as *timer_1*.



7. Select the *timer* instance and open the Interfaces tab. Now unfold the ***Memory Target*** and select *p_registers*. In the ***External Connections*** pane double click on *Unconnected (double-click to connect)*.



8. From the Add External Connection window select SharedMemoryMap and click on Connect



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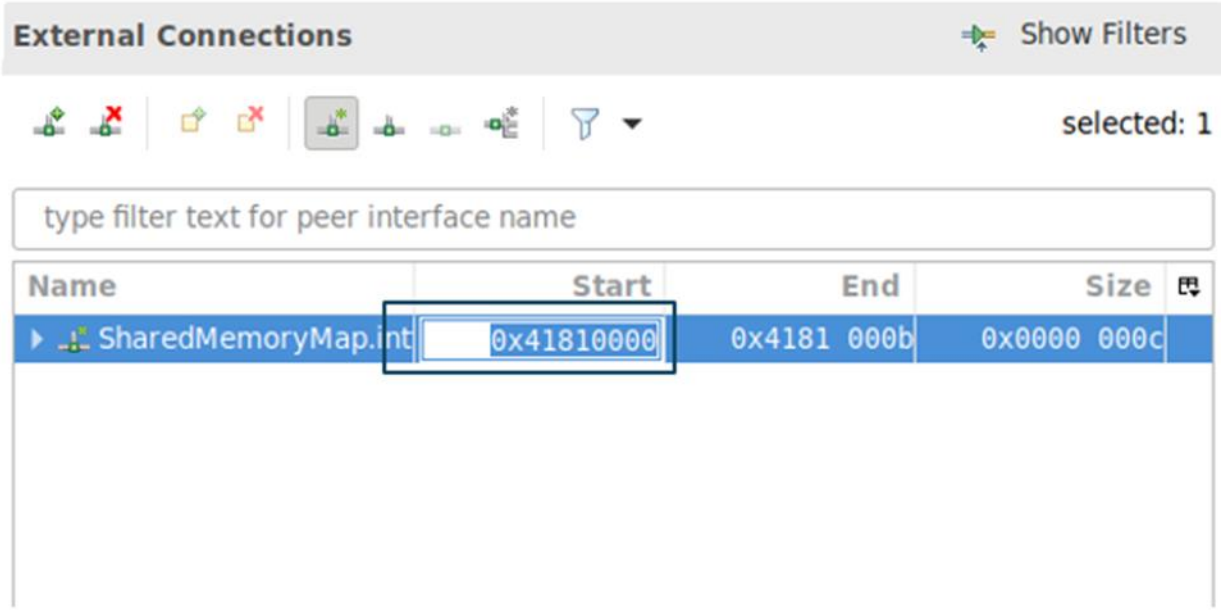
9. In the External Connections pane click on *SharedMemoryMap.int* and set Start to `0x41810000`.



Task-3: *Boot the System in the VDK Debug Perspective*

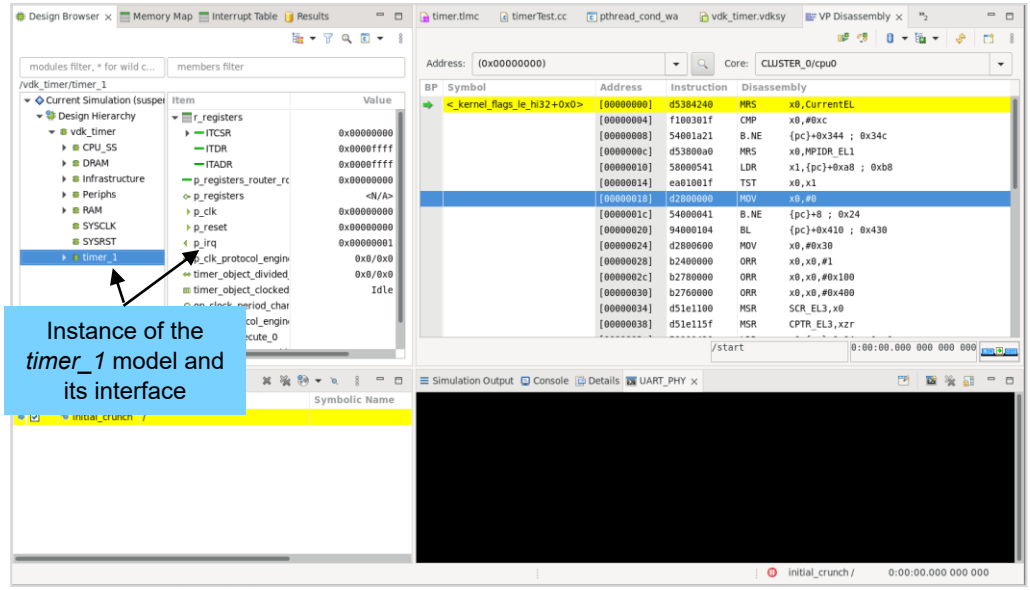
***Motivation**: We have integrated the timer model in our VDK, and we need to do a quick build & debug in the VDK Debug perspective of Virtualizer Studio to make sure that the model is connected and VDK project still runs as expected.*

10. Build the VDK project in the VDK Creation perspective by clicking the ***Build*** button.

    Check for errors, if any, in the ***Console*** view. You may ignore the warning related to the unconnected ports for the timer model.

11. Start the simulation by clicking the ***Run Simulation*** ( ) button.

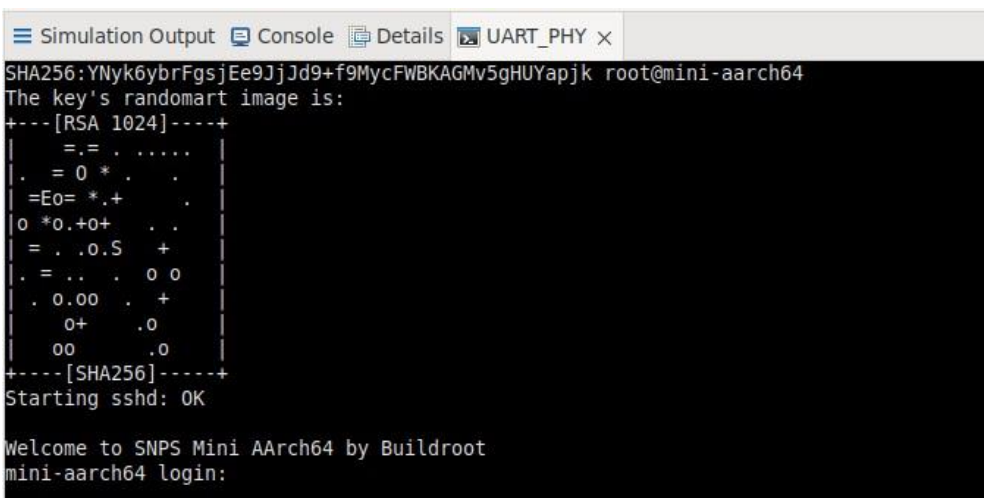
Virtualizer Studio automatically switches to the ***VDK Debug*** perspective and simulation pauses at `initial_crunch`.



12. Review the VDK components in the ***Design Browser*** to ensure that the timer model has been added. Also, check the simulation output messages in the ***Simulation Output*** view.

13. Click ***Resume*** ( ) to continue the simulation.

    Linux prompt appears in the ***UART_PHY*** window.



14. If done, stop the VDK execution. Click ***Stop*** ( ) to terminate the simulation in the VDK Debug perspective. Exit Virtualizer Studio using the ***File > Exit*** menu item.

What We Learnt

In this module, we performed the following tasks:

- Create a new VDK project in the ***VDK Creation*** perspective of Virtualizer Studio

- Open the timer Model Library in the ***Library Manager*** tab
- Add an instance of the model to your virtual prototype design.
- Observe the model in the system context in the ***VDK Debug*** Perspective.